■ Personalized Career Roadmap

Student → Data Scientist | Timeline: 6 Months | Built with Chain-of-Thought Reasoning

■ Current Position

Role: Student

Skills: Data Analysis

Strengths: Analytical thinking, data intuition, academic curiosity

Foundation: Strong base for DS transition — analysis is core DS work

■ Target Goal

Role: Junior / Entry-Level Data Scientist

Timeline: 6 Months

Avg Salary (India): ■6–12 LPA entry-level

Key Outcome: Portfolio + internship/job offer

■ Skill Gap Analysis

Skill Area	Current Level	Required Level	Gap	Priority
Python Programming	None / Beginner	Proficient	HIGH	■ Critical
Statistics & Probability	Basic (Analysis)	Strong	MEDIUM	■ High
Machine Learning	None	Intermediate	HIGH	■ Critical
SQL & Databases	Unknown	Proficient	MEDIUM	■ High
Data Visualization	Basic	Advanced	MEDIUM	■ Medium
Deep Learning / NLP	None	Awareness	LOW	■ Nice-to-have
Git & Dev Tools	None	Basic	LOW	■ Medium

■ Recommended Learning Plan

Month 1–2: Foundations	
• Python: NumPy, Pandas, Matplotlib	freeCodeCamp / Kaggle
• Statistics: Distributions, hypothesis testing	StatQuest, Khan Academy
• SQL: Joins, aggregations, window funcs	Mode Analytics / SQLZoo
Month 3–4: Core ML	
• Scikit-learn: Regression, Classification, Clustering	Hands-On ML (Géron)
• Feature Engineering & Model Evaluation	Kaggle Learn
• EDA & Visualization: Seaborn, Plotly	Towards Data Science
Month 5–6: Advanced & Job Prep	
• Deep Learning intro (TensorFlow/PyTorch basics)	fast.ai / Coursera
• MLOps basics: Docker, model deployment	DataTalks.Club
• DS Interview Prep: LeetCode (Easy/Med)	InterviewBit

■ Suggested Projects

1. EDA Capstone

Explore a real dataset (e.g. IPL / COVID India). Publish on Kaggle.

2. ML Classifier

Predict loan default or churn. Full pipeline: cleaning → model → eval.

3. End-to-End App

Deploy a price prediction API using Flask + Render. Share GitHub link.

4. Kaggle Competition

Participate in a beginner competition. Aim Top 30%. Add to resume.

5. Dashboard

Build a Tableau/Power BI or Streamlit dashboard on open data.

■ Networking Strategy

- **LinkedIn:** Post weekly learnings + project updates. Use #DataScience
- **Kaggle:** Engage in discussions, upvote notebooks, join teams
- **GitHub:** Keep all projects public. README-first approach
- **Communities:** Join Discord — Weights & Biases, DataTalks.Club
- **Outreach:** DM 3 DS professionals/week for 15-min chats
- **Events:** Attend local PyData / AI meetups (free on Meetup.com)
- **Twitter/X:** Follow @lexfridman, @hardmaru, @jeremyphoward

■ Monthly Milestones

Month	■ Milestone	■ Output
M 1	Python + Stats done	Cert / Notebook
M 2	SQL + EDA project	Kaggle Notebook
M 3	First ML model built	GitHub Repo
M 4	ML app deployed	Live URL
M 5	Kaggle competition entered	Leaderboard rank
M 6	Job applications sent	2+ interviews

■ Immediate Next Actions — Start TODAY

1	Install Python + Anaconda. Complete Python Basics on Kaggle Learn (free, 5 hrs).

2	Create a GitHub profile. Push your very first repository today (even a README).
3	Optimise LinkedIn: add 'Aspiring Data Scientist' headline + open-to-work badge.
4	Pick a dataset on Kaggle. Start an EDA notebook — just explore the data.
5	Set a daily 2-hour deep-work block in your calendar for the next 6 months.